

Problem 4

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Problem 1 Use the given information to find the equation of the tangent line.

- (a) Given $g(x) = x^3 f(x)$, $f(2) = 4$, and $f'(2) = 7$, find the equation of the tangent line to the graph of g at $x = 2$.
- (b) Given $h(z) = \frac{zs(z)}{z-3}$, $s(2) = 4$, and $s'(2) = 7$, find the equation of the tangent line to the graph of h at $z = 2$.
- (c) Given

x	1	2	3	4	5
$f(x)$	5	3	0	-4	3
$f'(x)$	-3	-5	-2	6	-4
$g(x)$	6	9	-8	13	15
$g'(x)$	8	5	-10	7	6

find the equation of the tangent line of

$$\frac{f(x)}{e^x g(x)}$$

at $x = 2$.